

Data Communication

Chapter-01

DATA COMMUNICATIONS

Telecommunication means communicating over a distance.

Data is any information, like numbers, words, or pictures, that can be used for analysis or sharing

Data communications involve exchanging this information between two devices (like computers or smartphones) using some kind of transmission medium. This medium can be things like wires, cables, or even wireless signals.

In short, data communications is all about sending and receiving information between devices so they can share data effectively.

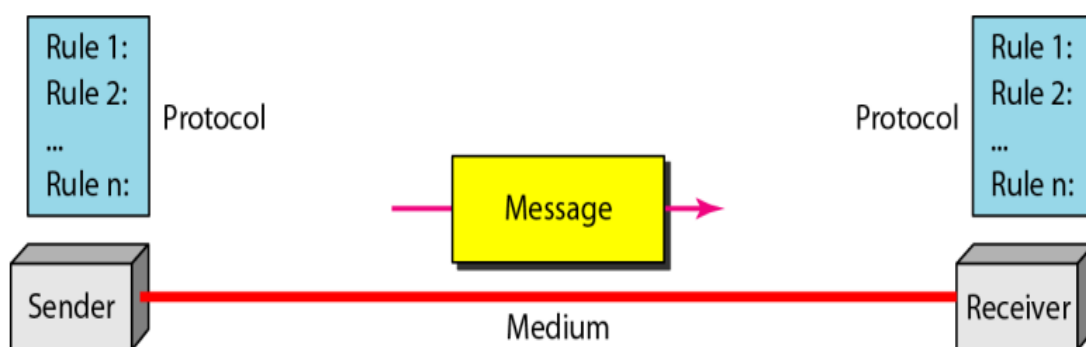
Five components of data communication:

1. Message

The **message** is the information being sent, like text, audio, or video.

2. Sender

The **sender** is the person or device (like a computer or smartphone) that creates and sends the message.



3. Receiver

The **receiver** is the person or device that gets the message.

4. Medium

The **medium** is the way the message travels, such as wires, cables, or wireless signals.

5. Protocol

Protocols are the rules that determine how the message is sent and received, ensuring everything works properly.

Data flow refers to the direction in which data moves between two devices during transmission. It describes how data is transmitted from the sender to the receiver.

Data flow

1. Simplex

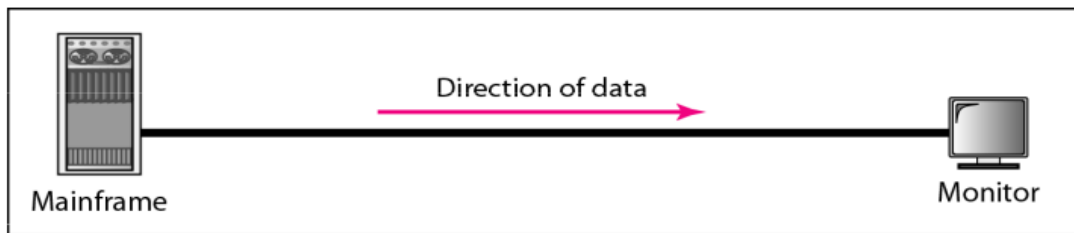
- **Definition:** Data flows in only one direction.
- **Example:** A keyboard sending data to a computer. The keyboard only sends input, and the computer does not send anything back.

2. Half-Duplex

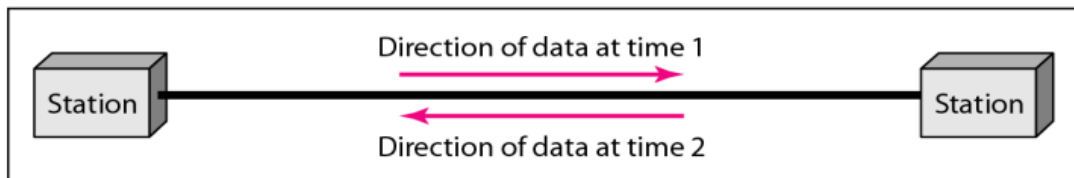
- **Definition:** Data can flow in both directions, but not at the same time.
- **Example:** A walkie-talkie. One person speaks while the other listens, and they must take turns to communicate.

3. Full-Duplex

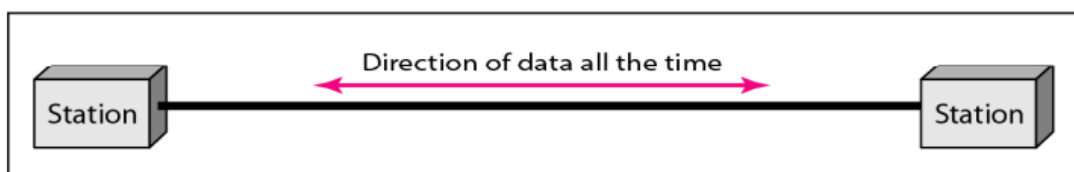
- **Definition:** Data can flow in both directions simultaneously.
- **Example:** A phone call. Both people can talk and listen at the same time without interrupting each other.



a. Simplex



b. Half-duplex



• c. Full-duplex

Network

A network is a group of devices, often called nodes, that are connected together through communication links. These devices can include computers, printers, or any other gadgets that can send or receive data. Each node can share information with other nodes on the network, allowing them to communicate and work together. This connection enables tasks like printing documents, sharing files, and accessing the internet.

Types of connections:

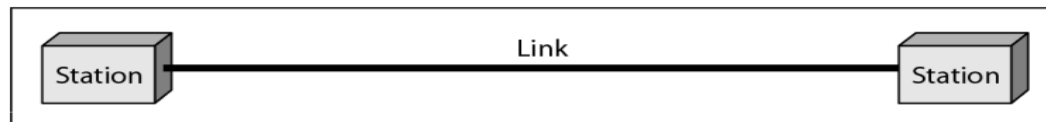
1. Point-to-Point Connection

- **Definition:** A direct link between two devices.
- **Example:** A connection between a computer and a printer. Data travels directly from the computer to the printer without any other devices involved.
- **Advantages:** Simple and faster since there is no competition for bandwidth.

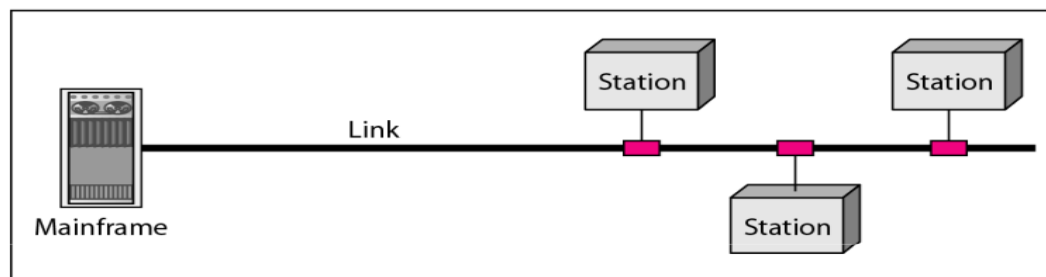
2. Multipoint Connection

- **Definition:** A single connection shared by multiple devices.

- **Example:** A network where several computers connect to a single server. All devices can communicate with the server, but they share the connection.
- **Advantages:** Cost-effective and efficient for connecting many devices, but may be slower due to shared bandwidth.

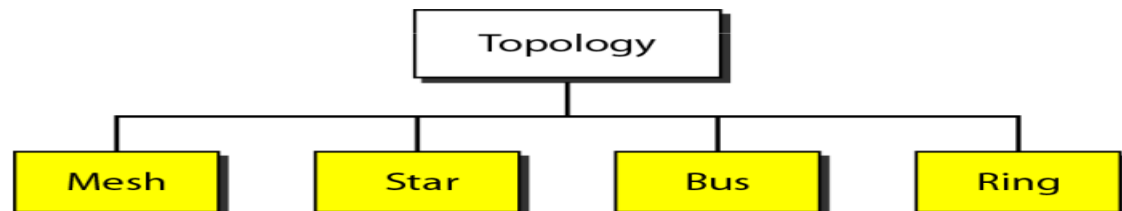


a. Point-to-point



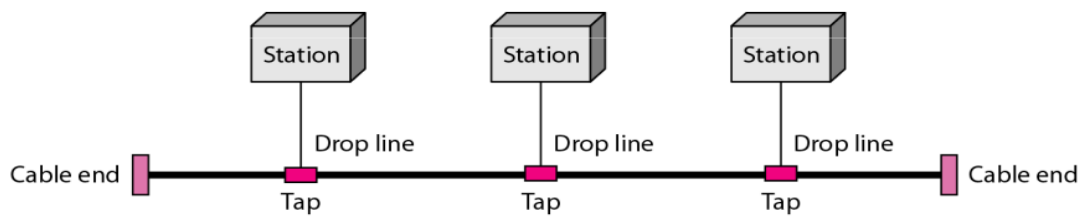
b. Multipoint

Categories of topology



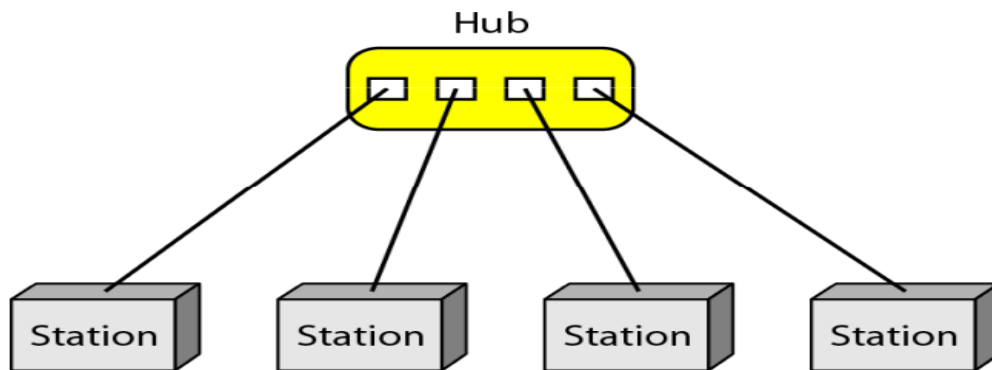
1. Bus Topology

- **Description:** All devices are connected to a single central cable, called the bus.
- **Advantages:** Simple and cost-effective for small networks.
- **Disadvantages:** If the central cable fails, the entire network goes down.



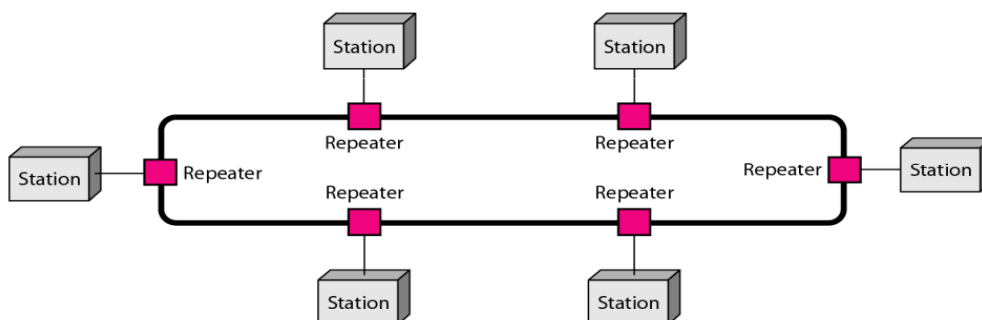
2. Star Topology

- **Description:** All devices are connected to a central hub or switch.
- **Advantages:** Easy to add or remove devices, and if one connection fails, it doesn't affect the rest of the network.
- **Disadvantages:** If the central hub fails, the whole network is affected.



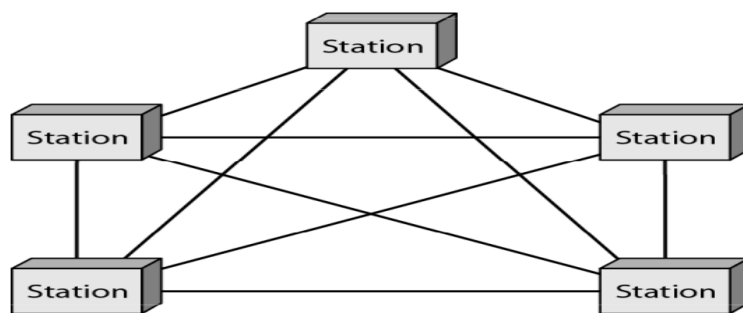
3. Ring Topology

- **Description:** Each device is connected to two other devices, forming a circular path for data.
- **Advantages:** Data travels in one direction, reducing the chance of data collisions.
- **Disadvantages:** If one device fails, it can disrupt the entire network.



4. Mesh Topology

- **Description:** Each device is connected to multiple other devices, creating multiple paths for data.
- **Advantages:** Highly reliable; if one connection fails, data can take another route.
- **Disadvantages:** More complex and expensive to set up due to many connections.



Hybrid topology

Hybrid topology combines two or more network topologies (like star, ring, or bus) for a flexible network design.

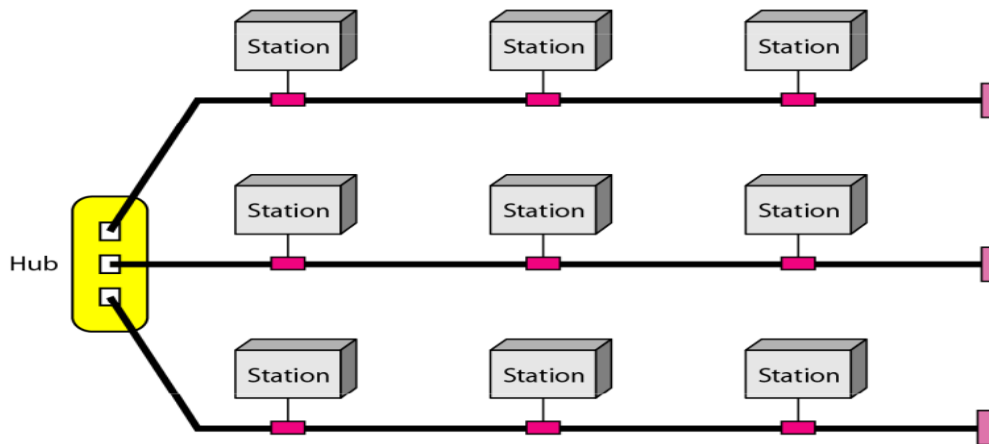
Advantages

- **Flexibility:** Adapts to changing needs.
- **Scalability:** Easy to expand.
- **Performance:** Optimizes data flow.
- **Reliability:** Parts can function independently if one fails.
- **Cost-Effective:** Utilizes existing infrastructure.

Disadvantages

- **Complexity:** More difficult to manage.
- **Cost:** Higher setup costs.
- **Maintenance:** Complicated troubleshooting.
- **Performance Risks:** Poor design can slow down the network.

A hybrid topology: a star backbone with three bus networks



Local Area Network (LAN)

- **Definition:** Connects devices in a small area (e.g., home, office).
- **Coverage:** Limited (up to a few kilometers).
- **Speed:** High data transfer speeds (up to 1 Gbps or more).
- **Ownership:** Typically owned by one organization.
- **Use:** Sharing resources like files and printers.

Wide Area Network (WAN)

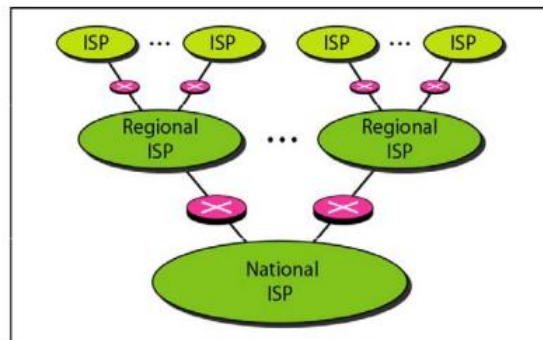
- **Definition:** Connects devices over large distances (e.g., cities, countries).
- **Coverage:** Broad (hundreds of kilometers to worldwide).
- **Speed:** Lower speeds compared to LANs (a few Mbps to several Gbps).
- **Ownership:** Often owned by multiple organizations or providers.
- **Use:** Connecting multiple LANs and providing internet access.

The Internet

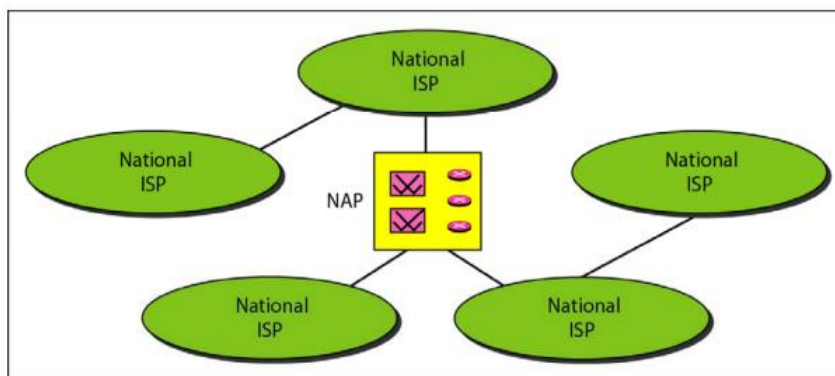
The **Internet** is a global network that connects millions of computers, allowing people to share information, communicate, and access resources. It enables activities like emailing, chatting, and online shopping. No single entity controls it; many organizations manage different parts. While it offers many benefits,

such as easy communication, it also faces challenges like security risks and unequal access for some users.

Hierarchical organization of the Internet



a. Structure of a national ISP



b. Interconnection of national ISPs

PROTOCOLS AND STANDARDS

Protocols are rules that govern how data is sent and received over a network, ensuring devices can communicate effectively. Examples include:

- **HTTP**: For web page transfers.
- **FTP**: For file sharing.
- **TCP/IP**: The main protocol suite for the Internet.

Standards are established norms or criteria that ensure compatibility and interoperability between devices and systems. Examples include:

- **IEEE 802.3**: For Ethernet networks.
- **IEEE 802.11**: For Wi-Fi.

In short, protocols facilitate data communication, while standards ensure different devices work together.

